

PRACTICAL WEEK -2

TASK 1:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid = fork();

    if (pid == 0)
    {
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());
        printf("Executing command: %s\n\n", command);

        execlp(command, command, NULL);

        printf("Command execution failed.\n");
    }
    else if (pid > 0)
    {
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);

        printf("\nChild process completed.\n");
    }
    else
    {
        printf("Fork failed!\n");
    }

    return 0;
}
```

```

(base) saiabburi@Geyasree-MacBook-Air OSSP % gcc command_executor.c -o command_executor
(base) saiabburi@Geyasree-MacBook-Air OSSP % ./command_executor
Enter a Linux command: ls

Parent Process
Parent PID : 7940
Child PID : 7941

Child Process
Child PID : 7941
Parent PID : 7940
Executing command: ls

LICENSE          assets          command_executor  fork_exec        obj              tests
Makefile         bin            command_executor.c  fork_exec.c      scripts
README.md        command_executor.c  docs              include          src

```

Child process completed.

```

(base) saiabburi@Geyasree-MacBook-Air OSSP %

```

TASK 2:

Uname -a :

```

Last login: Tue Aug 11 22:20:02 on ttys000
(base) saiabburi@Geyasree-MacBook-Air ~ % cd OSSP
(base) saiabburi@Geyasree-MacBook-Air OSSP % uname -a
Darwin Geyasree-MacBook-Air.local 22.5.0 Darwin Kernel Version 22.5.0: Thu Jun  8 22:21:34 PDT 2023; root:xnu-8796.121.3~7/RELEASE_ARM64_T8112 arm64

```

Lscpu :

Ps:

```

Darwin Geyasree-MacBook-Air.local 22.5.0 Darwin Kernel Version 22.5.0: Thu Jun  8 22:21:34 PDT 2023; root:xnu-8796.121.3~7/RELEASE_ARM64_T8112 arm64
(base) saiabburi@Geyasree-MacBook-Air OSSP % lscpu
[zsh: command not found: lscpu]
(base) saiabburi@Geyasree-MacBook-Air OSSP % ps
[  PID TTY          TIME CMD
  8285 ttys000      0:00.03 -zsh

```


Ps aux :

```
(base) saiabburi@Geyasree-MacBook-Air OSSP % ps aux
[USER] PID %CPU %MEM VSZ RSS TT STAT STARTED TIME COMMAND
saiabburi 7621 7.1 4.7 1922182880 393504 ?? S 10:08PM 2:28.12 /Applications/Go
_windowserver 372 1.6 0.8 410722144 63216 ?? Ss 8:28AM 77:40.12 /System/Library/
saiabburi 1445 0.9 2.4 1943926944 201632 ?? S 8:36AM 6:39.68 /Applications/Go
saiabburi 921 0.8 1.1 460556720 94928 ?? S 8:28AM 27:27.64 /Applications/Go
saiabburi 6568 0.8 4.7 1947840288 395248 ?? S 7:52PM 6:01.91 /Applications/Go
saiabburi 904 0.5 3.7 510055488 313152 ?? S 8:28AM 26:44.46 /Applications/Go
saiabburi 8299 0.5 1.6 19238810048 130176 ?? S 10:35PM 0:00.25 /Applications/Go
saiabburi 7873 0.4 0.6 409887456 51744 ?? S 10:22PM 0:06.32 /System/Applicat
saiabburi 7250 0.3 0.2 409241552 18096 ?? S 9:55PM 0:02.05 /System/Library/
saiabburi 7645 0.2 0.2 408284256 17152 ?? S 10:08PM 0:02.90 /System/Library/
root 342 0.1 0.1 408269024 9872 ?? Ss 8:28AM 0:25.63 /System/Library/
root 366 0.1 0.1 408264528 9104 ?? Ss 8:28AM 1:46.56 /usr/libexec/cor
saiabburi 925 0.1 1.0 459102208 85376 ?? S 8:28AM 17:19.24 /Applications/Go
saiabburi 7986 0.1 1.0 414513312 81696 ?? S 10:28PM 0:09.11 /Applications/Gr
saiabburi 6289 0.1 0.1 408103072 4368 ?? S 7:11PM 0:01.91 /System/Library/
saiabburi 7921 0.1 0.9 1923835712 74352 ?? S 10:24PM 0:01.23 /Applications/Go
_mysql 556 0.1 0.0 409623376 3888 ?? Ss 8:28AM 1:24.16 /usr/local/mysql
root 8221 0.0 0.1 408231360 6832 ?? Ss 10:32PM 0:00.11 /usr/libexec/sym
root 8220 0.0 0.1 408220176 5280 ?? Ss 10:32PM 0:00.19 /usr/bin/sysdiag
saiabburi 8218 0.0 0.1 408243680 12208 ?? S 10:32PM 0:00.30 /System/Library/
saiabburi 8217 0.0 0.1 408197200 5616 ?? S 10:32PM 0:00.07 /usr/libexec/pro
_accessoryupdater 8210 0.0 0.1 408226624 5552 ?? Ss 10:32PM 0:00.05 /System/Library/
saiabburi 8207 0.0 0.1 408207008 6688 ?? S 10:32PM 0:00.15 /usr/libexec/bio
root 8206 0.0 0.1 408228624 12336 ?? Ss 10:31PM 0:00.52 /usr/libexec/Pow
_accessoryupdater 8202 0.0 0.1 408226624 5200 ?? Ss 10:31PM 0:00.09 /System/Library/
saiabburi 8200 0.0 0.1 408243856 10176 ?? S 10:31PM 0:00.36 /System/Library/
saiabburi 8201 0.0 0.2 408229232 15328 ?? S 10:31PM 0:00.18 /System/Library/
saiabburi 8198 0.0 0.1 408685760 5072 ?? Ss 10:31PM 0:00.09 /System/Library/
```

Top :

```
(base) saiabburi@Geyasree-MacBook-Air OSSP % top
```

Processes: 392 total, 2 running, 390 sleeping, 2110 threads
Load Avg: 3.08, 2.16, 2.14 CPU usage: 1.83% user, 2.52% sys, 95.63% idle SharedLibs: 318M resident, 70M data, 14M linkedit. MemRegions: 308122 total, 2233M resident, 108M private, 976M shared.
PhysMem: 7538M used (1426M wired, 2599M compressor), 79M unused. VM: 168T vszize, 4283M framework vszize, 124450(0) swaptins, 250511(0) swaptouts. Networks: packets: 20513660/6801M in, 1462995/498M out.
Disks: 4902282/54G read, 4095819/34G written.

PID	COMMAND	%CPU	TIME	#TH	#WO	#PORTS	MEM	PURG	CMRPS	PGRP	PPID	STATE	BOOSTS	%CPU_ME	%CPU_OTHR	UID	FAULTS	COW	MSGSENT	MSGRECV	SYSSD	SYSMACH	CSW	PAGEIN	IDLEW	POWE	INSTRS
8330	top	7.5	00:08.56	1/1	0	29	8753K	0B	2656K-	8330	8285	running	*0[1]	0.00000	0.00000	0	41845+	72	1717214+	858601+	190043+	1198639+	4478+	29	2+	7.5	205208350
372	WindowServer	6.5	78:13.72	18	4	2091-	645M-	8032K-	129M	372	1	sleeping	*0[1]	0.05370	0.16669	88	7472364+	22308	31537656+	31288545+	51264535+	103088974+	23816549+	4088	138700+	6.6	70483861
0	kernel_task	6.1	68:58.52	464/8	0	0	15M	0B	0B	0	0	running	*0[0]	0.00000	0.00000	0	37920	0	46568692+	37274074+	245	0	293174598+	0	100104449+	0.0	36840928
6568	Google Chrom	2.6	06:15.43	37	1	459	739M-	0B	407M-	984	984	sleeping	*0[55]	0.00000	0.00000	501	2625142+	61222	1538741+	905953+	2459257+	3059609+	2118293+	592	15823+	2.8	14456179
8024	com.apple.We	1.6	00:20.38	6	2	88+	156M+	0B	40M-	8024	1	sleeping	*0[12266+]	0.00000	0.00000	501	76594+	180	1688304+	911169+	1047410+	4238925+	321461+	150	247	1.6	49482471
1445	Google Chrom	1.2	06:41.83	31	1	394	353M+	0B	256M-	984	984	sleeping	*0[43]	0.00000	0.00000	501	2314961+	30008	1562633+	492453+	1197484+	3163399+	2013219+	2831	6829	1.2	8596605
8021	app.inkwell	1.2	00:15.92	18	2	334+	29M+	0B	13M	8021	7986	sleeping	*0[89+]	0.00000	0.00000	501	53022+	253	258288+	108611+	590671+	1239458+	521703+	197	153	1.2	21307265
8027	com.apple.We	0.8	00:18.62	8	4	101+	21M+	48K	11M-	8027	1	sleeping	*0[31]	0.00000	0.00000	501	7654+	146	1690541+	2909466+	3158913+	7021466+	636408+	69	211	0.8	32908534
7873	Terminal	0.8	00:11.67	7	2	253	81M-	50M	15M	7873	1	sleeping	*0[149]	0.00572	0.00000	501	258732	321	94808+	24054+	132209+	263489+	79971+	817	35	0.8	58602512
8032	com.apple.We	0.7	00:11.13	4	1	80+	155M-	0B	96M-	8032	1	sleeping	*0[6368+]	0.00000	0.00000	501	70172+	172	1127799+	596080+	674359+	2758002+	215716+	4	171	0.7	16984258
7621	Google Chrom	0.6	02:32.64	21	1	234	403M	0B	363M-	984	984	sleeping	*0[21]	0.00000	0.00000	501	613416+	13250	330891	208220	511209	573881+	330633+	1455	795	0.6	4605388
8029	com.apple.We	0.5	00:01.65	5	2	81+	64M+	0B	21M-	8029	1	sleeping	*0[2812+]	0.00000	0.00000	501	28443+	173	21338+	12105+	27654+	69667+	24536+	8	29	0.5	9716595
8031	com.apple.We	0.5	00:04.33	7	2	90+	124M+	0B	39M-	8031	1	sleeping	*0[6112+]	0.00000	0.00000	501	48840+	182	68087+	39483+	67558+	216256+	46961+	174	68	0.5	11800762
8030	com.apple.We	0.5	00:01.95	5	2	81+	117M-	0B	62M-	8030	1	sleeping	*0[2943+]	0.00000	0.00000	501	33278+	173	31027+	17685+	34458+	91224+	26278+	20	37	0.5	7599057
904	Google Chrom	0.4	26:57.94	45	1	1290-	355M-	0B	192M-	984	1	sleeping	*0[7125]	0.00000	0.00000	501	7455632+	14740	28425197+	7672614+	42158467+	78628010+	17383252+	80653	10273	0.4	2517155
8337	screencaptur	0.4	00:01.11	2	1	66	7666K	752K	0B	622	622	sleeping	*0[995+]	0.01365	0.00000	501	15278+	111	14072+	4725+	4547+	26598+	6497+	101	0	0.4	3835322
7921	Google Chrom	0.4	00:01.59	18	1	131	58M-	0B	42M-	984	984	sleeping	*0[10]	0.00000	0.00000	501	23448+	876	11150+	9591+	20315+	25734+	15008+	24	6	0.4	2093209
556	mysqld	0.2	01:24.68	38	0	63	390M	0B	388M	556	1	sleeping	*0[1]	0.00000	0.00000	74	37038	106	115670+	57811+	8323419+	322336+	4128353+	1608	195267+	0.6	1821526
366	corebrihtne	0.1	01:46.83	3	2	129	3393K	0B	1392K	366	1	sleeping	*0[1]	0.16097	0.00000	0	169855+	0	422351+	286674+	2294006+	948308+	992417+	474	2241	0.1	1335921
925	Google Chrom	0.1	17:19.93	17	1	167	85M	0B	43M	984	984	sleeping	*0[3]	0.00000	0.00000	501	951176	1932	2309480+	1296658+	128156205+	4116723+	16283862+	5184	9125	0.1	505437
7986	Grammarly De	0.1	00:14.10	12	2	292	48M	144K	20M	7986	1	sleeping	*0[99+]	0.00000	0.00000	501	29650	668	170164+	26452+	2358003+	333461+	208128+	1673	188+	0.2	820481
7645	fileprovider	0.1	00:03.09	4	3	128	12M	0B	9696K	7645	1	sleeping	*0[357]	0.00000	0.00000	501	9772	218	3564	1290	75826+	7543	39919+	1242	160	0.1	608238
915	powerd	0.1	00:19.99	4	3	133	4193K	0B	1456K	315	1	sleeping	*0[1]	0.00000	0.05370	0	112805+	117	150877+	108393+	555391+	322229+	238496+	245	172	0.1	706529
557	com.muse.aut	0.0	00:19.01	51	2	26967+	5681K	0B	3952K	557	1	sleeping	*0[1]	0.00000	0.00000	0	11154	88	268832+	121032+	728446+	309157+	468345+	498	9388	0.0	424773
486	audioclocksy	0.0	00:15.70	4	2	63	4769K	0B	3072K	486	1	sleeping	*0[1]	0.00000	0.00000	0	18033	78	42153	245327+	228986+	478770+	240652+	93	14	0.0	253660
8338	screencaptur	0.0	00:00.17	4	2	147	7346K	0B	8338	1	sleeping	*0[836+]	0.00000	0.01365	501	2002	170	1516+	989+	5506+	5643+	2266+	116	0	0.0	254593	
6756	Google Chrom	0.0	16:53.51	21	1	287	732M	0B	662M	984	984	sleeping	*0[17]	0.00000	0.00000	501	2920532	23233	2335009	1802533	7659322	5174087+	6697187+	1754	10842	0.0	61796
2131	PerfPowerSer	0.0	01:50.30	5	3	323	13M	256K	3104K	2131	1	sleeping	*0[2179]	0.00000	0.00000	0	135408	160	1090089	916490+	1290789+	1574377	988368+	1470	179	0.0	162094
807	Google Drive	0.0	00:47.18	120	1	512	61M	0B	40M	780	780	sleeping	*0[1140]	0.00000	0.00000	501	373547	822	111688	34620	1446263+	413756	492152+	10549	9778	0.0	71462
7985	mobilitimer	0.0	00:00.08	3	2	56+	2545K+	0B	1536K	7985	1	sleeping	*0[1]	0.00000	0.00000	501	1690+	90	228+	57	1875+	613+	569+	101	2	0.0	133359
3475	AMPLibraryAg	0.0	00:03.17	5	1	130+	9089K+	0B	6320K-	3475	1	sleeping	*0[289]	0.00000	0.00000	501	283960+	223	4983	1924	38376+	18097	22332+	1305	154	0.0	147479
685	sharingd	0.0	00:33.50	4	1	321	15M	0B	13M	685	1	sleeping	*0[1]	0.00000	0.00000	501	102335	270	208783+	62267	658965+	756828+	328374+	2731	182	0.0	169649
302	lods	0.0	01:42.01	4	3	1272	26M	0B	26M	302	1	sleeping	*0[1]	0.00000	0.00000	0	276809	82	659360+	751838+	3512927+	750887+	1572665+	28193	256	0.0	232963
329	mgd	0.0	00:50.74	5	2	328-	38M-	0B	34M-	329	1	sleeping	*0[1]	0.00000	0.00000	0	226702+	133	58055	46187	1769395+	161107+	780488+	944	10033	0.0	118372
348	locationd	0.0	01:06.80	4	1	321	11M	0B	5024K-	348	1	sleeping	*0[31925]	0.00000	0.00000	205	138656+	171	81924	103444	151886+	391451+	545079+	14205	465	0.0	84863
7559	gamecontroll	0.0	00:00.12	3	2	46	3025K	0B	2064K	7559	1	sleeping	*0[1]	0.00000	0.00000	501	12917	96	519+	58	3463+	1868+	2315+	3	10	0.0	108716
699	lockoutagent	0.0	00:05.51	4	3	59	3041K	0B	1376K	699	1	sleeping	*0[1]	0.00000	0.00000	501	19894	90	30066+	2714	228843+	119908+	165977+	75	995	0.0	114855
339	thermalmonit	0.0	00:04.72	3	2	53	1937K	0B	960K	339	1	sleeping	*0[1]	0.00000	0.00000	0	6496	80	57529+	20762+	58380+	200940+	61182+	96	889	0.0	78427